

Chapter 4

The Mucosa at Teeth and Implants

Jan Lindhe, Jan L. Wennström, and Tord Berglundh

Department of Periodontology, Institute of Odontology, The Sahlgrenska Academy
at University of Gothenburg, Gothenburg, Sweden

Gingiva, 83	Quality, 90
Biologic width, 83	Vascular supply, 91
Dimensions of the buccal tissue, 83	Probing gingiva and peri-implant mucosa, 92
Dimensions of the interdental papilla, 84	Dimensions of the buccal soft tissue at implants, 94
Peri-implant mucosa, 85	Dimensions of the papilla between teeth and implants, 95
Biologic width, 86	Dimensions of the “papilla” between adjacent implants, 96

Gingiva

Biologic width

A term frequently used to describe the dimensions of the soft tissues that face the teeth is the *biologic width of the soft tissue attachment*. The development of the *biologic width concept* was based on studies and analyses by, among others, Gottlieb (1921), Orban and Köhler (1924), and Sicher (1959), who documented that the soft tissue attached to the teeth was comprised of two parts, one of fibrous tissue and one of epithelium. In a publication by Gargiulo *et al.* (1961) called “Dimensions and relations of the dentogingival junction in humans”, sections from autopsy block specimens that exhibited different degrees of “passive tooth eruption” (i.e. periodontal tissue breakdown) were examined. Histometric assessments were made to describe the length of the sulcus (not part of the attachment), the epithelial attachment (today called the junctional epithelium), and the connective tissue attachment (Fig. 4-1). It was observed that the length of the connective tissue attachment varied within narrow limits (1.06–1.08 mm), while the length of the attached epithelium was about 1.4 mm at sites with normal periodontium, 0.8 mm at sites with moderate and 0.7 mm at sites with advanced periodontal tissue breakdown. In other words, (1) the biologic width of the attachment varied between about 2.5 mm in the

normal case and 1.8 mm in the advanced disease case, and (2) the most variable part of the attachment was the length of the epithelial attachment (junctional epithelium).

Dimensions of the buccal tissue

The morphologic characteristics of the gingiva are related to the dimension of the alveolar process, the form (anatomy) of the teeth, events that occur during tooth eruption, and the eventual inclination and position of the fully erupted teeth (Wheeler 1961; O'Connor & Biggs 1964; Weisgold 1977). Oschenbein and Ross (1969) and Becker *et al.* (1997) proposed (1) that the anatomy of the gingiva is related to the contour of the osseous crest and (2) that two basic types of gingival architecture may exist, namely the “*pronounced scalloped*” and the “*flat*” biotype.

Subjects who belong to the “*pronounced scalloped*” biotype have long and slender teeth with tapered crown form, delicate cervical convexity, and minute interdental contact areas that are located close to the incisal edge (Fig. 4-2). The maxillary front teeth of such individuals are surrounded by a thin free gingiva, the buccal margin of which is located at or apical of the cemento-enamel junction. The zone of gingiva is narrow, and the outline of the gingival margin is highly scalloped (Olsson *et al.* 1993). On the

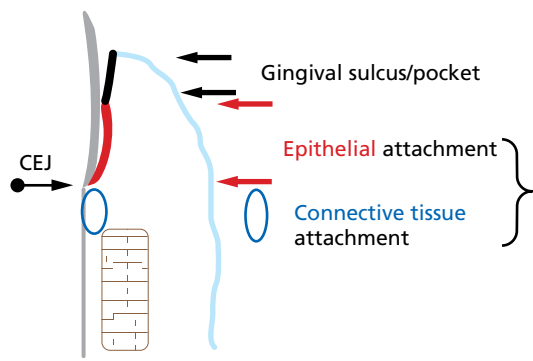


Fig. 4-1 Schematic drawing describing the “biologic width” of the soft tissue attachment at the buccal surface of a tooth with healthy periodontium. The combined length of the junctional epithelium (epithelial attachment) and the connective tissue attachment is considered to represent the “biologic width” of the soft tissue attachment. Note the gingival sulcus is *not* part of the attachment.

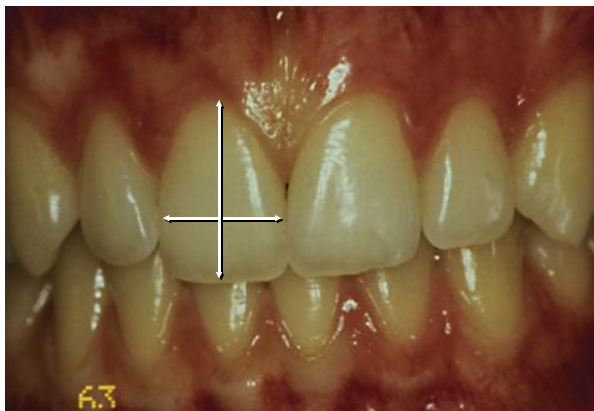


Fig. 4-2 Clinical photograph of a subject who belongs to the “pronounced scalloped” gingival biotype. The crowns of the teeth are comparatively long and slender. The papillae are comparatively long, the gingival margin is thin, and the zone of attached gingiva is short.

other hand, subjects who belong to the “flat” gingival biotype have incisors with squared crown form with pronounced cervical convexity (Fig. 4-3). The gingiva of such individuals is wider and more voluminous, the contact areas between the teeth are large and more apically located, and the interdental papillae are short. It was reported that subjects with a pronounced scalloped gingiva often exhibited more advanced soft tissue recession in the anterior maxilla than subjects with a flat gingiva (Olsson & Lindhe 1991).

Kan *et al.* (2003) measured the dimension of the gingiva – as determined by bone sounding – at the buccomesial and buccodistal aspects of maxillary anterior teeth. Bone sounding determines the distance between the soft tissue margin and the crest of the bone and, hence, provides an estimate that is about 1 mm greater than that obtained in a regular probing pocket depth measurement. The authors reported that the thickness of the gingiva varied between subjects of different gingival biotypes. Thus, the height of the gingiva at the buccal-approximal surfaces in subjects who belonged to the flat biotype was, on average, 4.5 mm, while in subjects belonging



Fig. 4-3 Clinical photograph of a subject who belongs to the “flat” gingival biotype. The crowns of the teeth are comparatively short but wide. The papillae are comparatively short but voluminous and the zone of attached gingiva is wide.

to the pronounced scalloped biotype the corresponding dimension (3.8 mm) was significantly smaller. This indicates that subjects who belong to the flat biotype have more voluminous soft buccal/approximal tissues than subjects who belong to the pronounced scalloped biotype.

Pontoriero and Carnevale (2001) evaluated the reformation of the gingival unit at the buccal aspect of teeth exposed to crown lengthening procedures using a denudation technique. At the 1-year follow-up examination after surgery, the regain of soft tissue – measured from the level of the denuded osseous crest – was greater in patients with a thick (flat) biotype than in those with a thin (pronounced scalloped) biotype (3.1 mm versus 2.5 mm). No assessment was made of the bone level change that had occurred between the baseline and the follow-up examination. It must, however, be anticipated that some bone resorption had taken place during healing and that the biologic width of the new connective tissue attachment had been re-established coronal to the level of the resected osseous crest.

The dimensions of the buccal gingiva may also be affected by the buccolingual position of the tooth within the alveolar process. A change of the tooth position in the buccal direction results in reduced dimensions of the buccal gingiva, while an increase is observed following a lingual tooth movement (Coatoam *et al.* 1981; Andlin-Sobocki & Brodin 1993). In fact, Müller and Könönen (2005) demonstrated in a study of the variability of the thickness of the buccal gingiva of young adults that most of the variation in gingival thickness was due to the tooth position and that the contribution of subject variability (i.e. flat and pronounced scalloped biotypes) was minimal.

Dimensions of the interdental papilla

The interdental papilla in a normal, healthy dentition has one buccal and one lingual/palatal component that are joined in the col region (see Chapter 1;

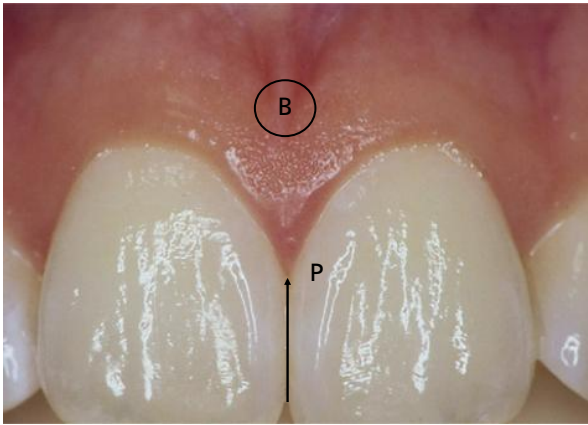


Fig. 4-4 Tarnow *et al.* (1992) measured the distance between the contact point (P) between the crowns of the teeth and the bone crest (B) using sounding (transgingival probing).

Figs. 1-1, 1-2, 1-3, 1-4, 1-5, 1-6, 1-7, 1-8, 1-9). Experiments performed in the 1960s (Kohl & Zander 1961; Matherson & Zander 1963) revealed that the shape of the papilla in the col region was not determined by the outline of the bone crest, but by the shape of the contact relationship that existed between adjacent teeth.

Tarnow *et al.* (1992) studied whether the distance between the contact point (area) between teeth and the crest of the corresponding interproximal bone could influence the degree of papilla fill that occurred at the site. Presence or absence of a papilla was determined visually in periodontally healthy subjects. If there was no space visible apical of the contact point, the papilla was considered complete. If a “black space” was visible at the site, the papilla was considered incomplete. The distance between the facial level of the contact point and the bone crest (Fig. 4-4) was measured by sounding. The measurement thus included not only the epithelium and connective tissue of the papilla, but in addition the entire supra-alveolar connective tissue in the interproximal area (Fig. 4-5). The authors reported that the papilla was always complete when the distance from the contact point to the crest of the bone was ≤ 5 mm. When this distance was 6 mm, papilla fill occurred in about 50% of cases and when ≥ 7 mm, it was incomplete in about 75% of cases. Considering that the supracrestal connective tissue attachment is about 1 mm high, these data indicate that the papilla height may be limited to about 4 mm in most cases. Interestingly, papillae of similar height (3.2–4.3 mm) were found to reform following surgical denudation procedures (van der Velden 1982; Pontoriero & Carnevale 2001), but to a greater height in patients with a thick (flat) than in those with a thin (pronounced scalloped) biotype.

Summary

- *Flat gingival (periodontal) biotype*: the buccal marginal gingiva is comparatively thick, the papillae are often short, the bone of the buccal cortical wall

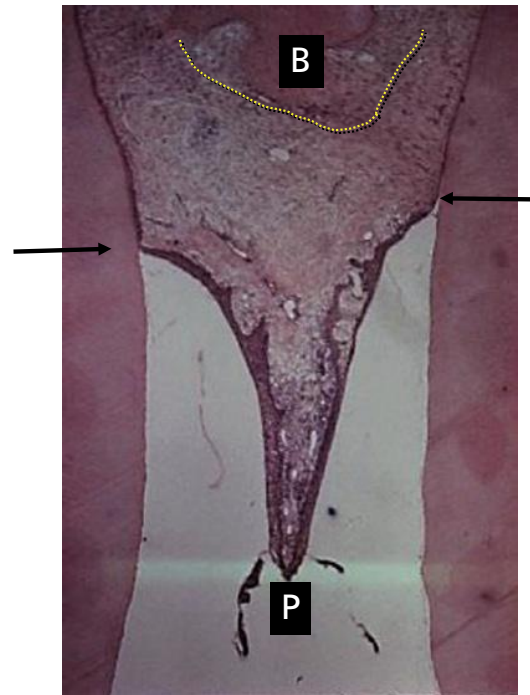


Fig. 4-5 Mesiodistal section of the interproximal area between the two central incisors. Arrows indicate the location of the cemento-enamel junction. Dotted line indicates the outline of the marginal bone crest. The distance between the contact point (P) between the crowns of the teeth and the bone crest (B) indicates the height of the papilla.

is thick, and the vertical distance between the interdental bone crest and the buccal bone is short (about 2 mm).

- *Pronounced scalloped gingival (periodontal) biotype*: the buccal marginal gingiva is delicate and may often be located apical of the cemento-enamel junction (receded), the papillae are high and slender, the buccal bone wall is often thin, and the vertical distance between the interdental bone crest and the buccal bone is long (>4 mm).

Peri-implant mucosa

The soft tissue that surrounds dental implants is termed *peri-implant mucosa*. Features of the peri-implant mucosa are established during the process of wound healing that occurs subsequent to the closure of mucoperiosteal flaps following implant installation (one-stage procedure) or following abutment connection (two-stage procedure) surgery. Healing of the mucosa results in the establishment of a soft tissue attachment (transmucosal attachment) to the implant. This attachment serves as a seal that prevents products from the oral cavity reaching the bone tissue, and thus ensures osseointegration and the rigid fixation of the implant.

The peri-implant mucosa and the gingiva have several clinical and histologic characteristics in common. Some important differences, however, also exist between the gingiva and the peri-implant mucosa.

Biologic width

The structure of the mucosa that surrounds implants made of titanium has been examined in humans and several animal models (for review, see Berglundh 1999). In an early study in the dog, Berglundh *et al.* (1991) compared some anatomic features of the gingiva at teeth and the mucosa at implants. Details of the research model used in this study are briefly outlined here, as this model was used in subsequent experiments that will be described in this chapter.

The mandibular premolars on one side of the mandible were extracted, leaving the corresponding teeth in the contralateral jaw quadrant. After 3 months of healing following tooth extraction (Fig. 4-6), the fixture part of the implants (Brånemark System®; Nobel Biocare, Gothenburg, Sweden) were installed (Fig. 4-7) and submerged according to the guidelines in the manual for the system. Another 3 months later, abutment connection was performed (Fig. 4-8) in a second-stage procedure, and the animals were placed in a carefully monitored plaque-control program. Four months subsequent to abutment connection, the dogs were exposed to a clinical examination following which biopsy specimens of several tooth and all implant sites were harvested.

The clinically healthy gingiva and peri-implant mucosa had a pink color and a firm consistency

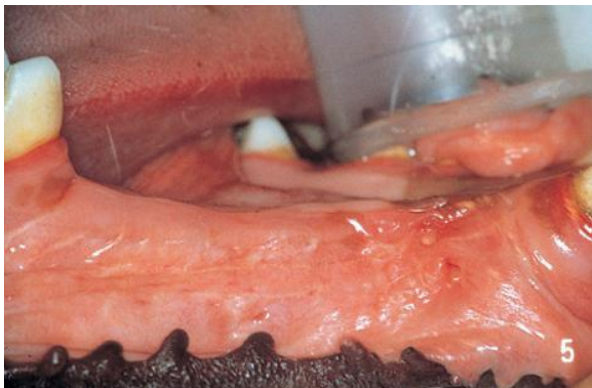


Fig. 4-6 Edentulous mandibular right premolar region 3 months following tooth extraction. (Source: Berglundh *et al.* 1991. Reproduced with permission from John Wiley & Sons.)

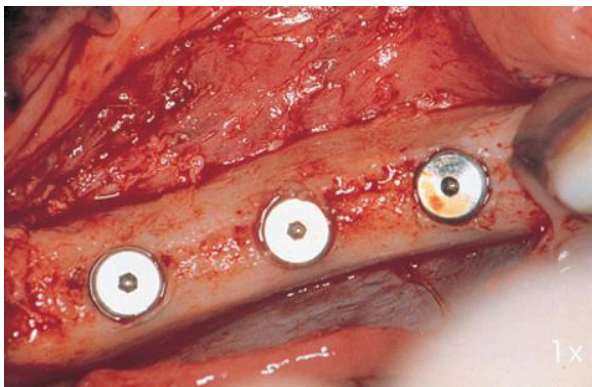


Fig. 4-7 Three titanium implants (i.e. the fixture part and cover screw; Brånemark System®) are installed.

(Fig. 4-9). On radiographs obtained from the tooth sites, it was observed that the alveolar bone crest was located about 1 mm apical of a line connecting the cemento-enamel junction of neighboring premolars (Fig. 4-10). The radiographs from the implant sites disclosed that the bone crest was close to the junction between the abutment and the fixture part of the implant (Fig. 4-11).

Histologic examination of the sections revealed that the two soft tissue units, the gingiva and the peri-implant mucosa, had several features in common. The oral epithelium of the gingiva was well

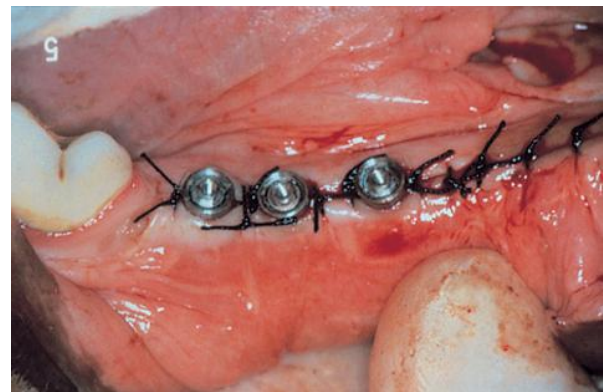


Fig. 4-8 Abutment connection is performed and the mucosa sutured with interrupted sutures.

(a)



(b)



Fig. 4-9 After 4 months of careful plaque control, the gingiva (a) and the peri-implant mucosa (b) are clinically healthy.



Fig. 4-10 Radiograph obtained of the premolars in the left side of the mandible.

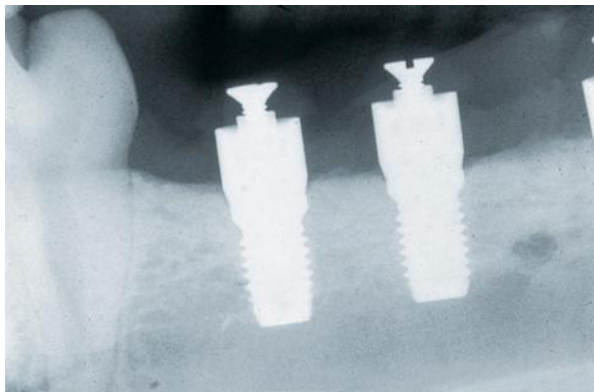


Fig. 4-11 Radiograph obtained of the implants in the right side of the mandible.

keratinized and continuous with the thin junctional epithelium that faced the enamel and that ended at the cementoenamel junction (Fig. 4-12). The supra-alveolar connective tissue was about 1 mm high and the periodontal ligament about 0.2–0.3 mm wide. The principal fibers were observed to extend from the root cementum in a fan-shaped pattern into the soft and hard tissues of the marginal periodontium (Fig. 4-13).

The outer surface of the peri-implant mucosa was also covered by a keratinized oral epithelium, which in the marginal border connected with a thin barrier epithelium (similar to the junctional epithelium at the teeth) that faced the abutment part of the implant (Fig. 4-14). It was observed that the barrier epithelium was only a few cell layers thick (Fig. 4-15) and that the epithelial structure terminated about 2 mm apical of the soft tissue margin (Fig. 4-14) and 1–1.5 mm from the bone crest. The connective tissue in the compartment above the bone appeared to be in direct contact with the surface (TiO_2) of the implant (Figs. 4-14, 4-15, 4-16). The collagen fibers in this connective tissue apparently originated from the periosteum of the bone crest and extended towards the margin of the soft tissue in directions parallel to the surface of the abutment.

The observation that the barrier epithelium of the healthy mucosa consistently ended at a certain

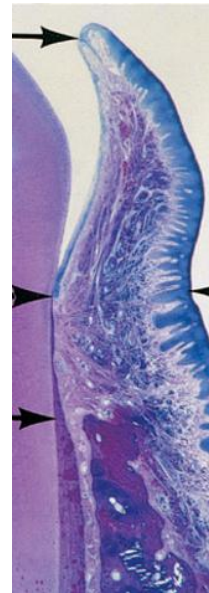


Fig. 4-12 Microphotograph of a cross-section of the buccal and coronal part of the periodontium of a mandibular premolar. Note the position of the soft tissue margin (top arrow), the apical cells of the junctional epithelium (center arrowheads), and the crest of the alveolar bone (bottom arrow). The junctional epithelium is about 2 mm long and the supracrestal connective tissue portion about 1 mm high.

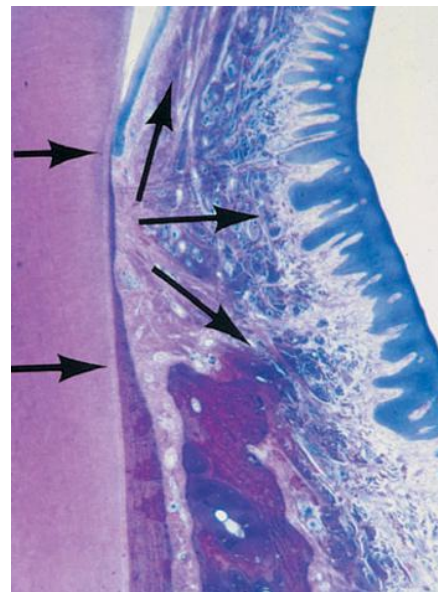


Fig. 4-13 Higher magnification of the supracrestal connective tissue portion seen in Fig. 4-12. Note the direction of the principal fibers (arrows).

distance (1–1.5 mm) from the bone is important. During healing following implant installation surgery, fibroblasts of the connective tissue of the mucosa apparently formed a biologic attachment to the TiO_2 layer of the “apical” portion of the abutment portion of the implant. This attachment zone was evidently not recognized as a wound and was therefore not covered with an epithelial lining.

In further dog experiments (Abrahamsson *et al.* 1996, 2002), it was observed that a similar mucosal attachment formed when different types of implant

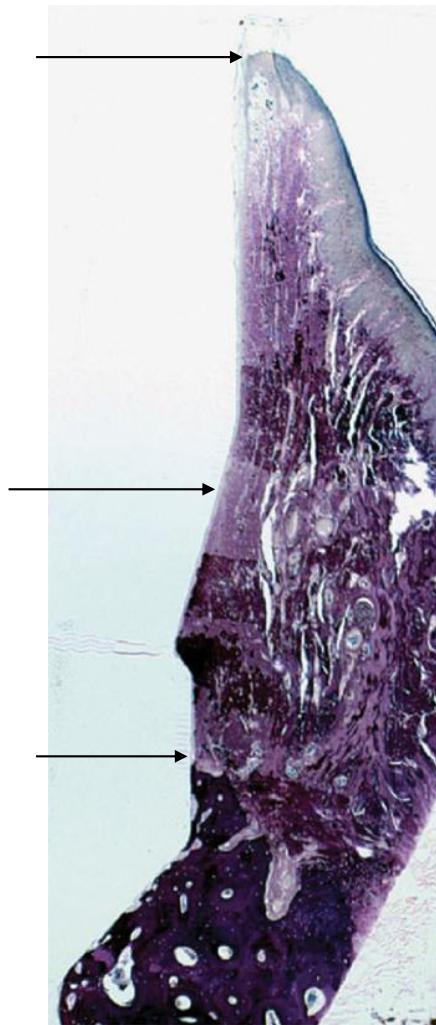


Fig. 4-14 Microphotograph of a buccolingual section of the peri-implant mucosa. Note the position of the soft tissue margin (top arrow), the apical cells of the junctional epithelium (center arrow), and the crest of the marginal bone (bottom arrow). The junctional epithelium is about 2 mm long and the implant-connective tissue interface about 1.5 mm high.

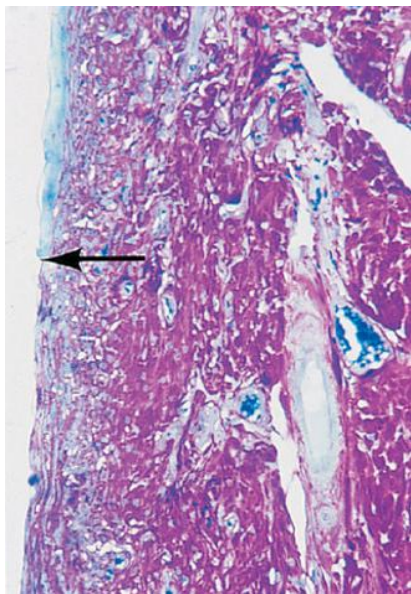


Fig. 4-15 Higher magnification of the apical portion of the barrier epithelium (arrow) in Fig. 4-14.

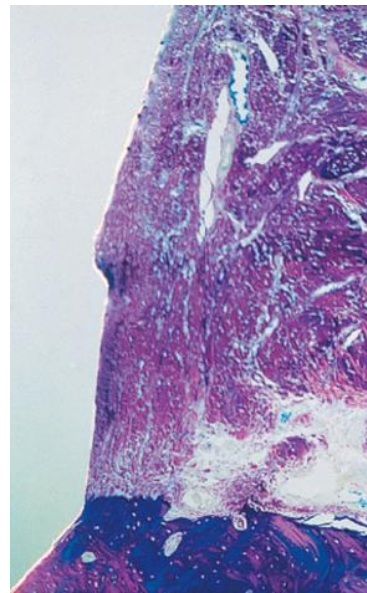


Fig. 4-16 Microphotograph of a section (buccolingual) of the implant-connective tissue interface of the peri-implant mucosa. The collagen fibers invest in the periosteum of the bone and project in directions parallel to the implant surface towards the margin of the soft tissue.



Fig. 4-17 Implants of three systems installed in the mandible of a Beagle dog. AstraTech Implants® Dental System (left), Brånemark System® (center) and ITI® Dental Implant System (right).

systems were used (e.g. AstraTech Implant System, AstraTech Dental, Mölndal, Sweden; Brånemark System®; Straumann® Dental Implant System, Straumann AG, Basel, Switzerland; 3i® Implant System, Implant Innovation Inc., West Palm Beach, FL, USA). In addition, the formation of the attachment appeared to be independent of whether the implants were initially submerged or not (Figs. 4-17, 4-18).

In subsequent studies (Abrahamsson *et al.* 1998; Welander *et al.* 2008), it was demonstrated that the material used in the abutment part of the implant was of decisive importance for the location of the connective tissue portion of the transmucosal attachment. Abutments made of aluminum-based sintered ceramic (Al_2O_3) and zirconium dioxide (ZrO_2) allowed for the establishment of a mucosal attachment similar to that which occurred at titanium abutments. Abutments made of a gold alloy or dental

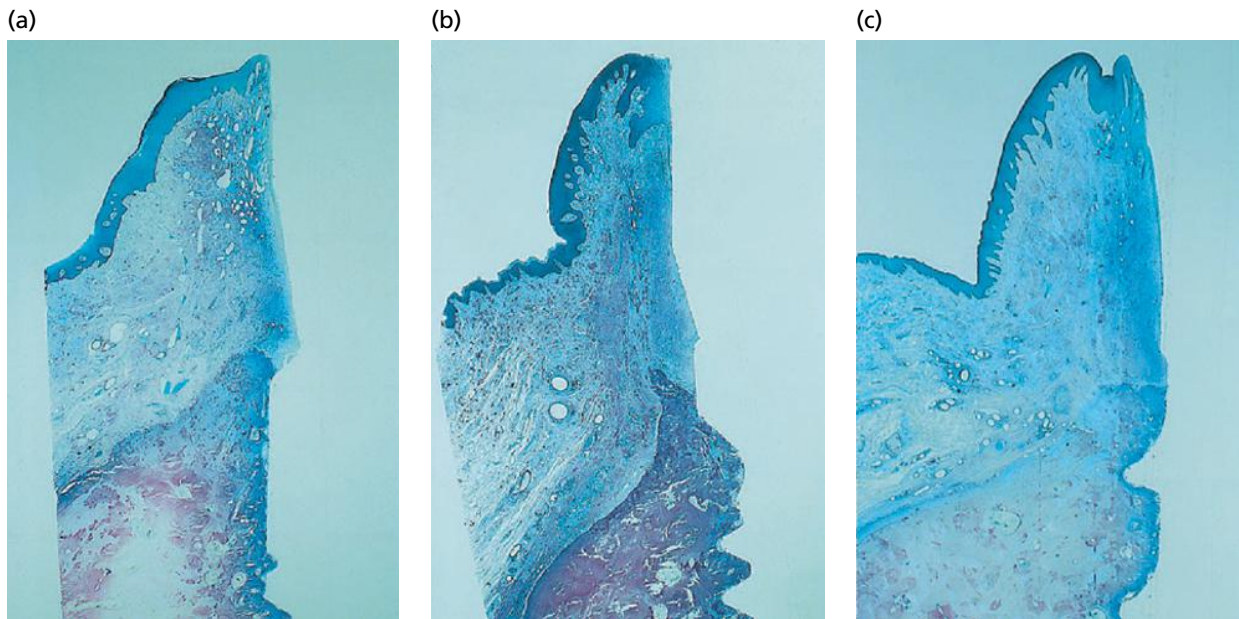


Fig. 4-18 Microphotographs illustrating the mucosa (buccolingual view) facing the three implant systems. (a) AstraTech Implants® Dental System. (b) Brånemark System®. (c) ITI® Dental Implant System.

porcelain, however, provided conditions for inferior mucosal healing. When such materials were used, the connective tissue attachment failed to develop at the abutment level. Instead, the connective tissue attachment occurred in a more apical location. Thus, during healing following the abutment connection surgery, some resorption of the marginal peri-implant bone took place to expose the titanium portion of the fixture (Brånemark System®) to which the connective tissue attachment eventually formed.

The location and dimensions of the transmucosal attachment were examined in a dog experiment by Berglundh and Lindhe (1996). Implants (fixtures) of the Brånemark System® were installed in edentulous premolar sites and submerged. After 3 months of healing, abutment connection was performed. On the left side of the mandible, the volume of the ridge mucosa was maintained, while on the right side the vertical dimension of the mucosa was reduced to 2 mm or less (Fig. 4-19) before the flaps were replaced and sutured. In biopsy specimens obtained after another 6 months, it was observed that the transmucosal attachment at all implants included a barrier epithelium component that was about 2 mm long and a zone of connective tissue that was about 1.3–1.8 mm high.

Further examination disclosed that at sites with a thin mucosa, wound healing had consistently included marginal bone resorption to establish space for a mucosa that eventually could harbor both the epithelial and the connective tissue components of the transmucosal attachment (Figs. 4-20, 4-21).

The dimensions of the epithelial and connective tissue components of the transmucosal attachment at implants are established during wound healing following implant surgery. As is the case for bone healing after implant placement (see Chapter 5),

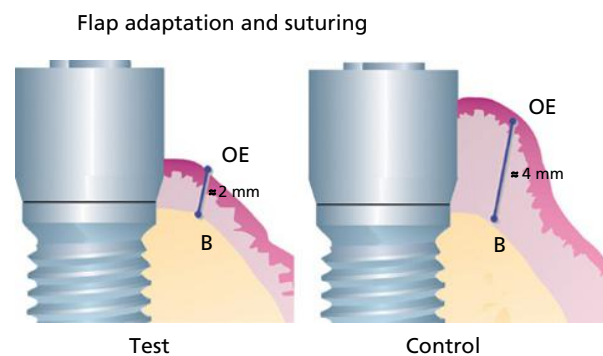


Fig. 4-19 Schematic drawing illustrating that the mucosa at the test site was reduced to about 2 mm. (Source: Berglundh *et al.* 1991. Reproduced with permission from John Wiley & Sons.)

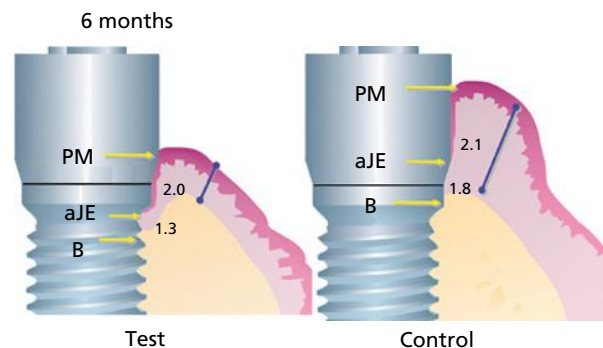


Fig. 4-20 Schematic drawing illustrating that the peri-implant mucosa at both control and test sites contained a 2-mm long barrier epithelium and a zone of connective tissue that was about 1.3–1.8 mm high. Bone resorption occurred in order to accommodate the soft tissue attachment at sites with a thin mucosa. (Source: Berglundh & Lindhe 1996. Reproduced with permission from John Wiley & Sons.)

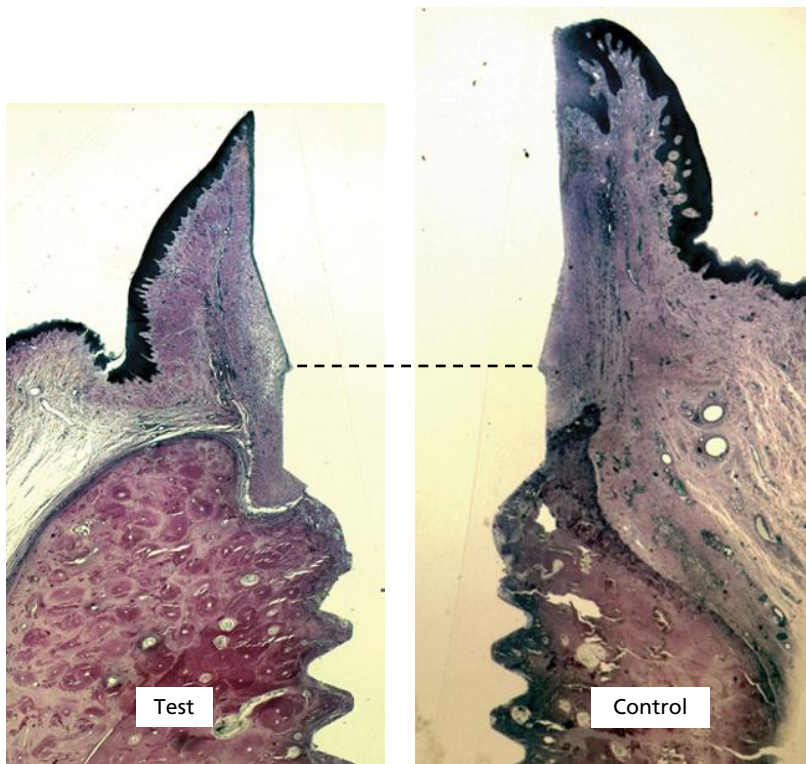


Fig. 4-21 Microphotograph illustrating the peri-implant mucosa of a normal dimension (Control) and reduced dimension (Test). Note the angular bone loss that had occurred at the site with the thin mucosa.

the wound healing in the mucosa around implants is a delicate process that requires several weeks of tissue remodeling.

In an animal experiment, Berglundh *et al.* (2007) described the morphogenesis of the mucosa attachment to implants made of c.p. titanium. A non-submerged implant installation technique was used and the mucosal tissues were secured to the conical marginal portion of the implants (Straumann® Dental Implant System) with interrupted sutures. The sutures were removed after 2 weeks and a plaque-control program was initiated. Biopsies were performed at various intervals to provide healing periods extending from day 0 (2 hours) to 12 weeks. It was reported that large numbers of neutrophils infiltrated and degraded the coagulum that occupied the compartment between the mucosa and the implant during the initial phase of healing. The first signs of epithelial proliferation were observed in specimens representing 1–2 weeks of healing and a mature barrier epithelium was seen after 6–8 weeks. It was also demonstrated that the collagen fibers of the mucosa were organized after 4–6 weeks of healing. Thus, prior to this time interval, the connective tissue is not properly arranged.

Summary: The junctional and barrier epithelia are about 2 mm long and the zones of supra-alveolar connective tissue are between 1 and 1.5 mm high. Both epithelia are attached via hemi-desmosomes to the tooth/implant surface (Gould *et al.* 1984). The main attachment fibers (the principal fibers) invest in the root cementum of the tooth, but at the implant site the equivalent fibers run in a direction parallel to the implant and fail to attach to

the metal body. The soft tissue attachment to implants is properly established several weeks following surgery.

Quality

The quality of the connective tissue in the supra-alveolar compartments at teeth and implants was examined by Berglundh *et al.* (1991). The authors observed that the main difference between the mesenchymal tissue present at a tooth and at an implant site was the occurrence of cementum on the root surface in the former. From this cementum (Fig. 4-22), coarse dentogingival and dentoalveolar collagen fiber bundles projected in lateral, coronal, and apical directions (see Fig. 4-13). At the implant site, the collagen fiber bundles were orientated in an entirely different manner. Thus, the fibers invested in the periosteum at the bone crest and projected in directions parallel to the implant surface (Fig. 4-23). Some of the fibers became aligned as coarse bundles in areas distant from the implant (Buser *et al.* 1992).

The connective tissue in the supracrestal area at implants was found to contain more collagen fibers, but fewer fibroblasts and vascular structures, than the tissue in the corresponding location at teeth. Moon *et al.* (1999), in a dog experiment, reported that the attachment tissue close to the implant (Fig. 4-24) contained only a few blood vessels, but a large number of fibroblasts that were orientated with their long axes parallel to the implant surface (Fig. 4-25). In more lateral compartments, there were fewer fibroblasts, but more collagen fibers and

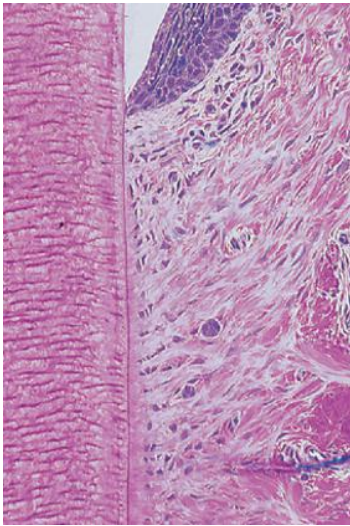


Fig. 4-22 Microphotograph of a tooth with marginal periodontal tissues (buccolingual section). Note on the tooth side the presence of an acellular root cementum with inserting collagen fibers. The fibers are orientated more or less perpendicular to the root surface.

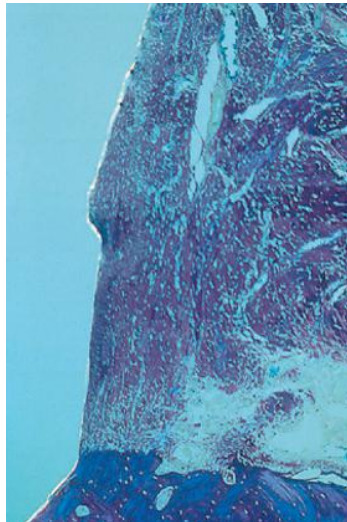


Fig. 4-23 Microphotograph of the peri-implant mucosa and the bone at the tissue-titanium interface. Note that the orientation of the collagen fibers is more or less parallel (not perpendicular) to the titanium surface.

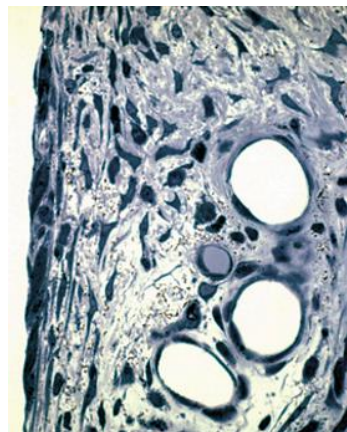


Fig. 4-24 Microphotograph of the implant-connective tissue interface of the peri-implant mucosa. A large number of fibroblasts reside in the tissue next to the implant.



Fig. 4-25 Electron micrograph of the implant-connective tissue interface. Elongated fibroblasts are interposed between thin collagen fibrils (magnification $\times 24\,000$).



Fig. 4-26 Buccolingual cleared section of a Beagle dog gingiva. The vessels have been filled with carbon (arrows). Note the presence of a suprapariosteal vessel on the outside of the alveolar bone, the presence of a plexus of vessels within the periodontal ligament, as well as vascular structures in the very marginal portion of the gingiva.

more vascular structures. From these and other similar findings, it may be concluded that the connective tissue attachment between the titanium surface and the connective tissue is established and maintained by fibroblasts.

Vascular supply

The vascular supply to the gingiva comes from two different sources (Fig. 4-26). The first source is represented by the large *suprapariosteal blood vessels* that put forth branches to form (1) the capillaries of the connective tissue papillae under the oral epithelium and (2) the vascular plexus lateral to the junctional epithelium. The second source is the *vascular*

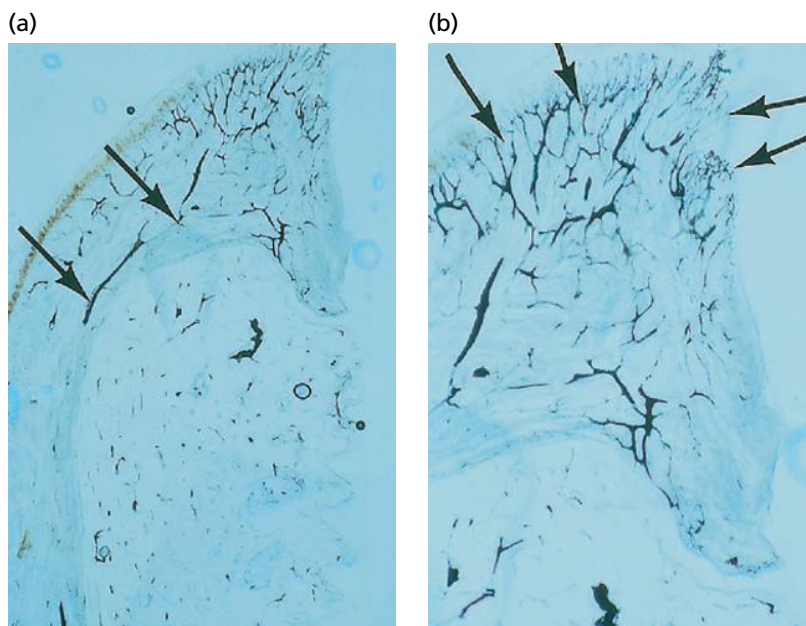


Fig. 4-27 (a) Buccolingual cleared section of a Beagle dog mucosa facing an implant (the implant was positioned to the right). Note the presence of a suprapariosteal vessel on the outside of the alveolar bone (arrows), but also that there is no vasculature that corresponds to the periodontal ligament plexus. (b) Higher magnification (of a) of the peri-implant soft tissue and the bone implant interface. Note the presence of a vascular plexus lateral to the junctional epithelium (arrows), but the absence of vessels in the more apical portions of the soft tissue facing the implant and the bone.

plexus of the periodontal ligament, from which branches run in a coronal direction and terminate in the supra-alveolar portion of the free gingiva. Thus, the blood supply to the zone of supra-alveolar connective tissue attachment in the periodontium is derived from two apparently independent sources (see Chapter 1).

Berglundh *et al.* (1994) observed that the vascular system of the peri-implant mucosa of dogs (Fig. 4-27) originated *solely* from the large *suprapariosteal blood vessel* on the outside of the alveolar ridge. This vessel gave off branches to the supra-alveolar mucosa and formed (1) the capillaries beneath the oral epithelium and (2) the vascular plexus located immediately lateral to the barrier epithelium. The connective tissue part of the transmucosal attachment to titanium implants contained only a few vessels, all of which could be identified as terminal branches of the *suprapariosteal blood vessels*.

Summary: The gingiva at teeth and the mucosa at dental implants share some characteristics, but differ in the composition of the connective tissue, the alignment of the collagen fiber bundles, and the distribution of vascular structures in the compartment apical of the barrier epithelium.

Probing gingiva and peri-implant mucosa

It was assumed for many years that the tip of the probe in a pocket depth measurement identified the most apical cells of the junctional (pocket) epithelium or the marginal level of the connective tissue attachment. This assumption was based on findings by, for example, Waerhaug (1952), who reported that the “epithelial attachment” (e.g. Gottlieb 1921; Orban & Köhler 1924) offered no resistance to probing. Waerhaug (1952) inserted, “with the greatest caution”,

thin blades of steel or acrylic into the gingival pocket of various teeth of >100 young subjects without signs of periodontal pathology. In several sites the blades were placed in approximal pockets, “in which position radiograms were taken of them”. It was concluded that the insertion of the blades could be performed without resulting in bleeding and that the device consistently reached the cemento-enamel junction (Fig. 4-28). Thus, the epithelium or the epithelial attachment offered no resistance to the insertion of the device.

In subsequent studies it was observed, however, that the tip of a periodontal probe in a pocket depth measurement only identified the base of the dentogingival epithelium by chance. In the absence of an inflammatory lesion, the probe frequently failed to reach the apical part of the junctional epithelium (e.g. Armitage *et al.* 1977; Magnusson & Listgarten 1980). If an inflammatory lesion, rich in leukocytes and poor in collagen, was present in the gingival connective tissue, however, the probe penetrated beyond the epithelium to reach the apicolateral border of the infiltrate.

Probing depth measurements at *implant sites* was examined in various animal models. Ericsson and Lindhe (1993) used the model of Berglundh *et al.* (1991) referred to earlier and, hence, had both teeth and implants available for examination. The gingiva at mandibular premolars and the mucosa at correspondingly positioned implants (Brånemark System®) were, after extended periods of plaque control, considered clinically healthy. A probe with a tip diameter of 0.5 mm was inserted into the buccal “pocket” using a standardized force of 0.5 N. The probe was anchored to the tooth or to the implant and biopsies from the various sites were performed. The histologic examination of the biopsy material revealed that probing

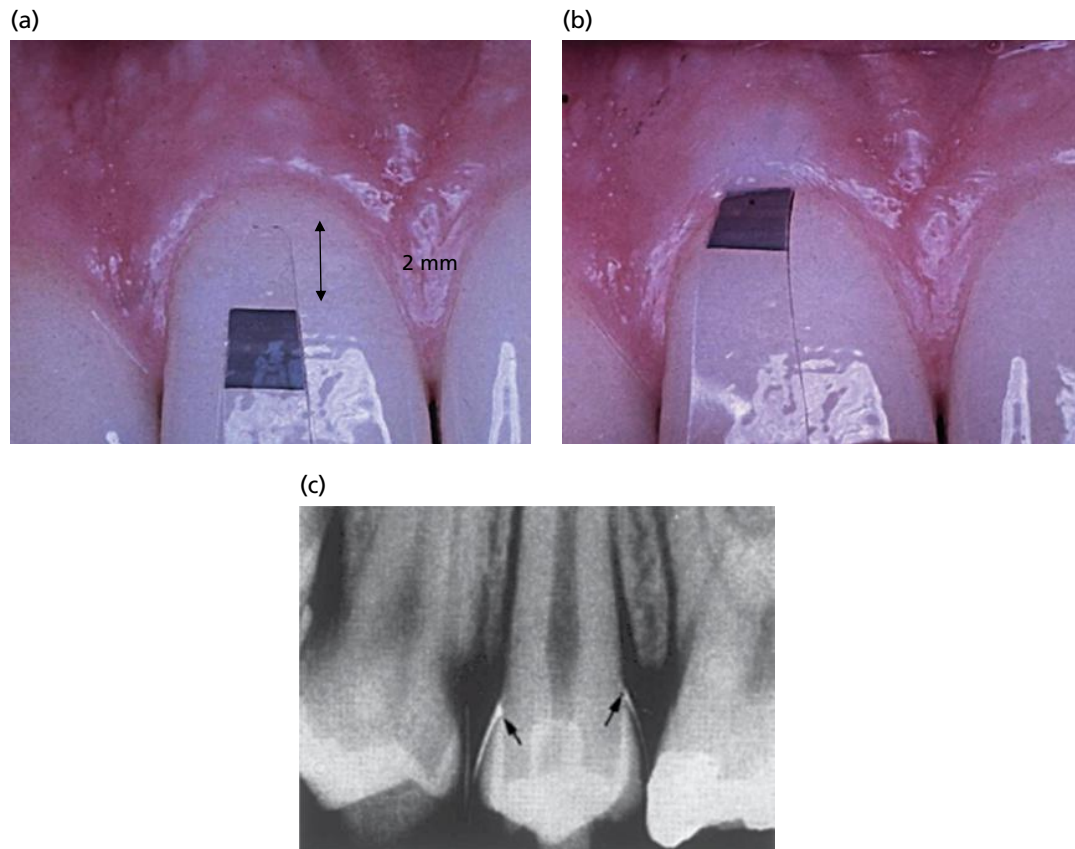


Fig. 4-28 Acrylic strip with a blue zone located 2 mm from the strip margin (a) prior to and (b) after its insertion into a buccal “pocket”. With a light force the strip could be inserted 2 mm into the “pocket”. (c) Thin blades of steel were inserted into pockets at approximal sites of teeth with healthy periodontal tissue. On radiographs, Waerhaug (1952) could observe that the blades consistently reached the cementoenamel junction.

the dentogingival interface had resulted in a slight compression of the gingival tissue. The tip of the probe was located coronal to the apical cells of the junctional epithelium. At the implant sites, probing caused both compression and a lateral dislocation of the peri-implant mucosa, and the average “histologic” probing depth was markedly deeper than at the tooth site: 2.0 mm versus 0.7 mm. The tip of the probe was consistently positioned deep in the connective tissue–abutment interface and apical of the barrier epithelium. The distance between the probe tip and the bone crest at the tooth sites was about 1.2 mm. The corresponding distance at the implant site was 0.2 mm. The findings presented by Ericsson and Lindhe (1993) regarding the difference in probe penetration in healthy gingiva and peri-implant mucosa are not in agreement with data reported in subsequent animal experiments.

Lang *et al.* (1994) used Beagle dogs and prepared the implant (Straumann® Dental Implant System) sites in such a way that on probing some were healthy, a few exhibited signs of mucositis, and some exhibited peri-implantitis. Probes with different geometry were inserted into the pockets using a standardized probing procedure and a force of only 0.2 N. The probes were anchored and block biopsy specimens were harvested. The probe locations were studied in

histologic ground sections. The authors reported that the mean “histologic” probing depth at healthy sites was about 1.8 mm, that is similar to the depth (about 2 mm) recorded by Ericsson and Lindhe (1993). The corresponding depth at sites with mucositis and peri-implantitis was about 1.6 mm and 3.8 mm, respectively. Lang *et al.* (1994) further stated that at healthy and mucositis sites, the probe tip identified “the connective tissue adhesion level” (i.e. the base of the barrier epithelium), while at peri-implantitis sites, the probe exceeded the base of the ulcerated pocket epithelium by a mean distance of 0.5 mm. At such peri-implantitis sites, the probe reached the base of the inflammatory cell infiltrate.

Schou *et al.* (2002) compared probing measurements at implants and teeth in eight cynomolgus monkeys. Ground sections were produced from tooth and implant sites that were (1) clinically healthy, (2) slightly inflamed (mucositis/gingivitis), and (3) severely inflamed (peri-implantitis/periodontitis) and in which probes had been inserted. An electronic probe (Peri-Probe®) with a tip diameter 0.5 mm and a standardized probing force of 0.3–0.4 N was used. It was demonstrated that the probe tip was located at a similar distance from the bone in healthy tooth sites and implant sites. On the other hand, at implants exhibiting mucositis and peri-implantitis, the probe

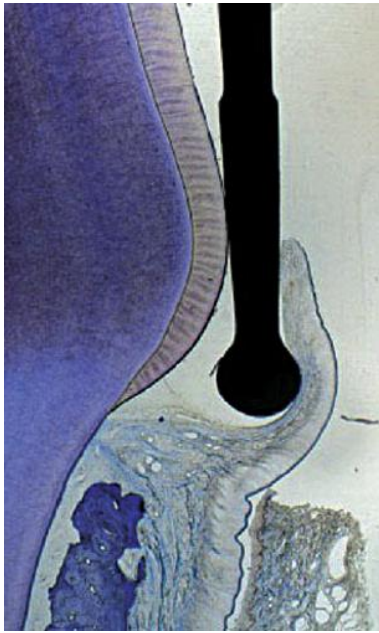


Fig. 4-29 Buccolingual ground section from a tooth site illustrating the probe tip position in relation to the bone crest. (Source: Abrahamsson & Soldini 2006. Reproduced with permission from John Wiley & Sons.).

tip was consistently identified at a more apical position than at corresponding tooth sites (gingivitis and periodontitis). The authors concluded that (1) probing depth measurements at implant and teeth yielded different information, and (2) small alterations in probing depth at implants may reflect changes in soft tissue inflammation rather than loss of supporting tissues.

Abrahamsson and Soldini (2006) evaluated the location of the probe tip in healthy periodontal and peri-implant tissues in dogs. It was reported that probing with a force of 0.2N resulted in a probe penetration that was similar at implants and teeth. Furthermore, the tip of the probe was often at or close to the apical cells of the junctional/barrier epithelium. The distance between the tip of the probe and the bone crest was about 1mm at both teeth and implants (Figs. 4-29, 4-30). Similar observations were reported from clinical studies in which different implant systems were used (Buser *et al.* 1990; Quirynen *et al.* 1991; Mombelli *et al.* 1997). In these studies, the distance between the probe tip and the bone was assessed in radiographs and was found to vary between 0.75 and 1.4mm when a probing force of 0.25–0.45N was used.

By comparing the findings from the studies reported above, it becomes apparent that probing depth and probing attachment level measurements are also meaningful at implant sites. When a “normal” probing force is applied to healthy tissues, the probe seems to reach similar levels at implant and tooth sites. Probing inflamed tissues at both tooth and implant sites will, however, result in a more advanced probe penetration and the tip of the probe may come closer to the bone crest.



Fig. 4-30 Buccolingual ground section from an implant site illustrating the probe tip position in relation to the bone crest. (Source: Abrahamsson & Soldini 2006. Reproduced with permission from John Wiley & Sons.)

Dimensions of the buccal soft tissue at implants

Chang *et al.* (1999) compared the dimensions of the periodontal and peri-implant soft tissues of 20 subjects who had been treated with an implant-supported single-tooth restoration in the esthetic zone of the maxilla and who had a non-restored natural tooth in the contralateral position (Fig. 4-31). In comparison to the natural tooth, the implant-supported crown was bordered by a thicker buccal mucosa (2.0mm versus 1.1mm), as assessed at a level corresponding to the bottom of the probeable pocket, and had a greater probing pocket depth (2.9mm versus 2.5mm) (Fig. 4-32). It was further observed that the soft tissue margin at the implant was more apically located (about 1mm) than the gingival margin at the contralateral tooth.

Kan *et al.* (2003) studied the dimensions of the peri-implant mucosa at 45 single implants that had been placed in the anterior maxilla for an average of 33 months. Bone sounding measurements performed at the buccal aspect of the implants showed that the height of the mucosa was 3–4mm in the majority of the cases. Less than 3mm of mucosa height was found at only 9% of the implants. It was suggested that implants in this category (1) were found in subjects who belonged to a *thin periodontal biotype*, (2) had been placed too labially, and/or (3) had the emergence of an over-contoured facial prosthetic. A peri-implant soft tissue dimension of >4mm was usually associated with a *thick periodontal biotype*.

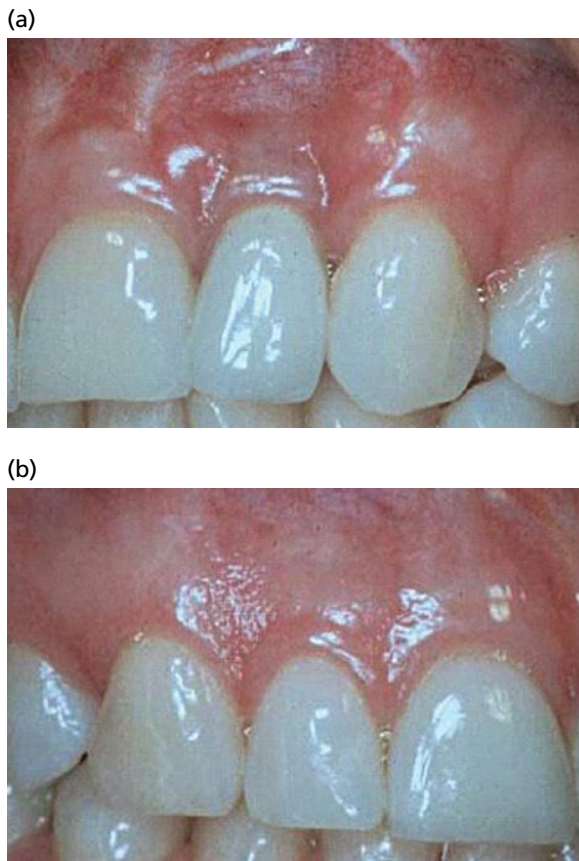


Fig. 4-31 Clinical photographs of (a) an implant-supported single-tooth replacement in position 12 and (b) the natural tooth in the contralateral position. (Source: Chang *et al.* 1999. Reproduced with permission from John Wiley & Sons.)

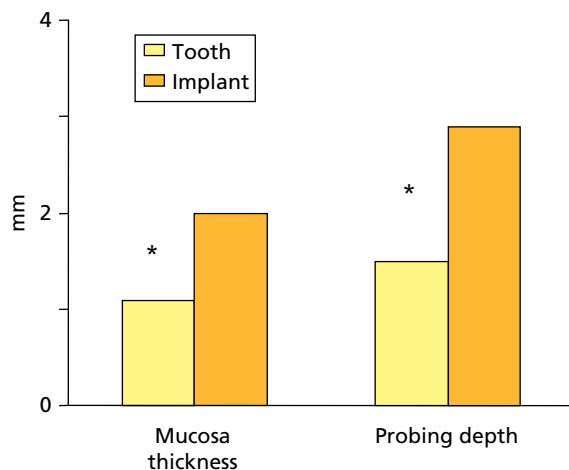


Fig. 4-32 Comparison of mucosa thickness and probing depth at the facial aspect of single-implant restorations and the natural tooth in the contralateral position. (Source: Chang *et al.* 1999. Reproduced with permission from John Wiley & Sons.)

Dimensions of the papilla between teeth and implants

In a study by Schropp *et al.* (2003), it was demonstrated that following single-tooth extraction the height of the papilla at the adjacent teeth was reduced by about 1mm. Concomitant with this reduction (recession) of the papilla height, the pocket depth

was reduced and some loss of clinical attachment occurred.

Following single-tooth extraction and subsequent implant installation, the height of the papilla in the tooth-implant site will be dependent on the attachment level of the tooth. Choquet *et al.* (2001) studied the papilla level adjacent to single-tooth dental implants in 26 patients and a total of 27 implant sites. The distance between the apical extension of the contact point between the crowns and the bone crest, as well as the distance between the soft tissue level and the bone crest, was measured on radiographs. The examinations were made 6–75 months after the insertion of the crown restoration. The authors observed that the papilla height consistently was about 4mm and, depending on the location of the contact point between adjacent crown papilla, fill was either complete or incomplete (Fig. 4-33). The closer the contact point was to the incisal edge of the crowns (restorations), the less complete was the papilla fill.

Chang *et al.* (1999) studied the dimensions of the papillae at implant-supported single-tooth restorations in the anterior region of the maxilla and at non-restored contralateral natural teeth. They found that the papilla height at the implant-supported crown was significantly shorter and showed less fill of the embrasure space than the papilla at the natural tooth (Fig. 4-34). This was particularly evident for the distal papillae of implant-supported restorations in the central incisor position, both in comparison to the distal papilla at the contralateral tooth and to the papilla at the mesial aspect of the implant crown. This indicates that the anatomy of the adjacent natural teeth (e.g. the diameter of the root, the proximal outline/curvature of the cemento-enamel junction/connective tissue attachment level) may have a profound influence on the dimension of the papilla lateral to an implant. Hence, the wider faciolingual root diameter and the higher proximal curvature of the cemento-enamel junction of the maxillary central incisor – in comparison to corresponding dimensions of the lateral incisor (Wheeler 1966) – may favor the maintenance of the height of the mesial papilla at the single-implant-supported restoration.

Kan *et al.* (2003) assessed the dimensions of the peri-implant mucosa lateral to 45 single implants placed in the anterior maxilla and the 90 adjacent teeth using bone sounding measurements. The bone sounding measurements were performed at the mesial and distal aspects of the implants and at the mesial and distal aspects of the teeth. The authors reported that the thickness of the mucosa at the mesial/distal surfaces of the implant sites was on average 6mm, while the corresponding dimension at the adjacent tooth sites was about 4mm. It was further observed that the dimensions of the peri-implant mucosa of subjects who belonged to the *thick periodontal biotype* were significantly greater than those of subjects with a *thin biotype*.

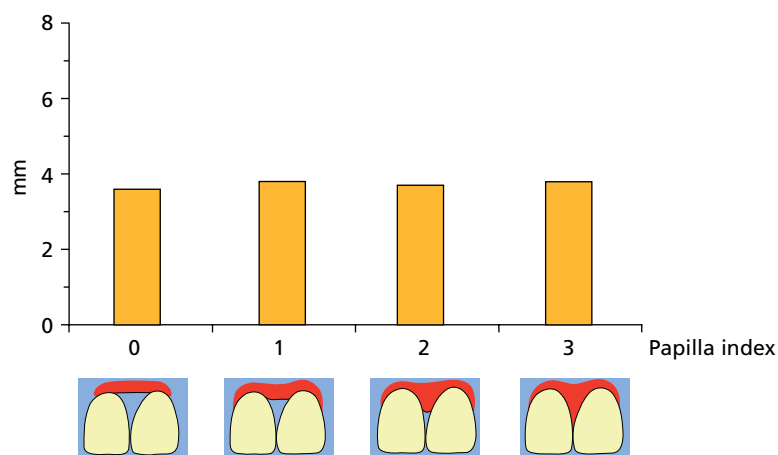


Fig. 4-33 Soft tissue height adjacent to single-tooth dental implants in relation to the degree of papilla fill. (Source: Choquet *et al.* 2001. Reproduced from the American Academy of Periodontology.)

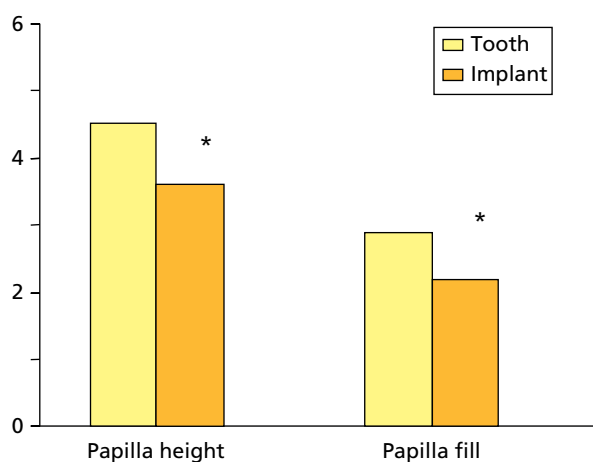


Fig. 4-34 Comparison of papilla height and papilla fill adjacent to single-implant restorations and the natural tooth in the contralateral position. (Source: Chang *et al.* 1999. Reproduced with permission from John Wiley & Sons.)

The level of the connective tissue attachment on the adjacent tooth surface and the position of the contact point between the crowns are obviously key factors that determine whether or not a complete papilla fill will be obtained at the single-tooth implant-supported restoration (Fig. 4-35). Although there are indications that the dimensions of the approximal soft tissue may vary between individuals having thin and thick periodontal biotypes, the height of the papilla at the single-implant restoration seems to have a biologic limit of about 4 mm (compare this with the dimension of the interdental papilla). Hence, to achieve a complete papilla fill of the embrasure space, a correct location of the contact area between the implant crown and the tooth crown is mandatory. In this respect it must also be recognized that the papilla fill at single-tooth implant restorations is unrelated to whether the implant is inserted according to a one- or two-stage protocol and whether a crown restoration is inserted immediately following surgery or delayed until the soft tissues have healed (Jemt 1999; Ryser *et al.* 2005).

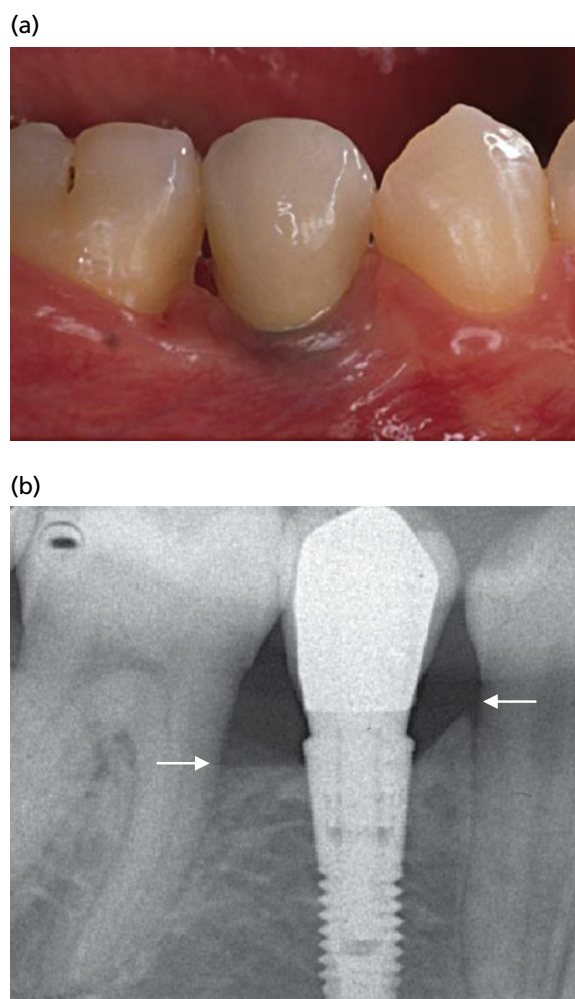


Fig. 4-35 Single implant in a mandibular premolar region. (a) Papilla fill between the implant and the first premolar is optimal, while the papilla fill between the implant and the molar is compromised and a black space is visible. (b) Radiograph from the same site showing the position of the cemento-enamel junction (on the premolar) and the marginal bone level (on the molar) (arrows).

Dimensions of the “papilla” between adjacent implants

When two neighboring teeth are extracted, the papilla at the site will be lost (Fig. 4-36). Hence, at replacement of the extracted teeth with implant-supported

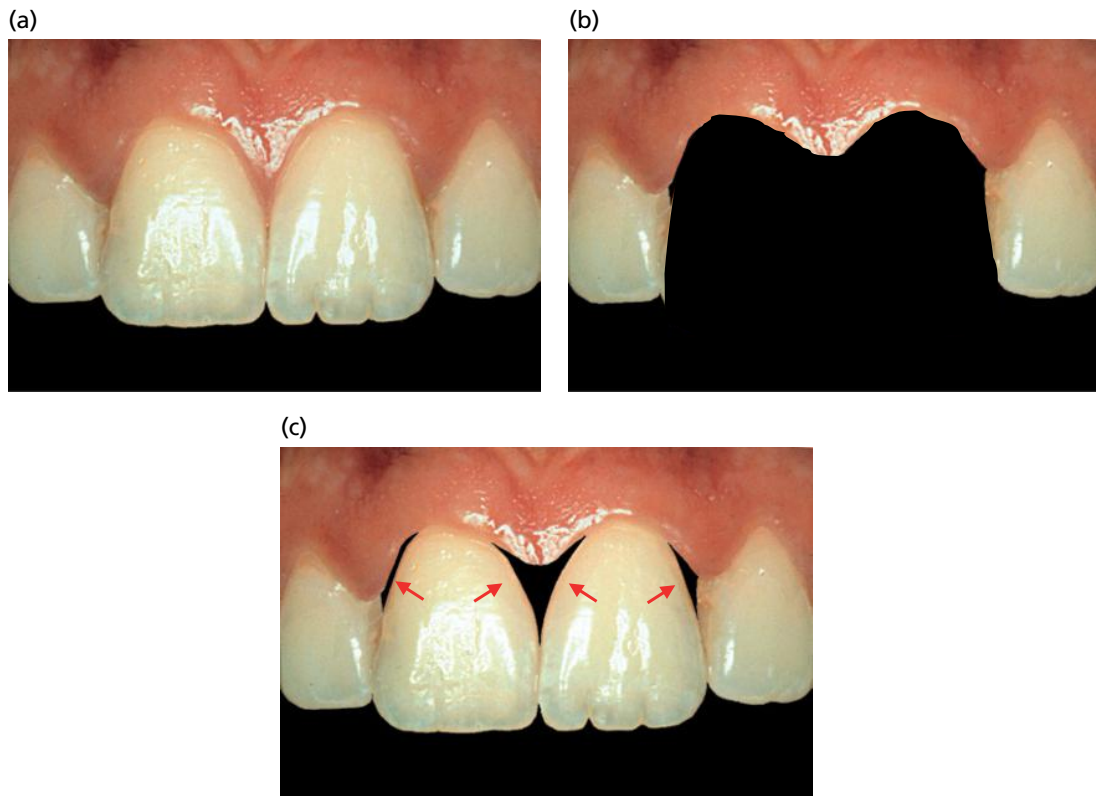


Fig. 4-36 See text for details. Arrows indicate the position of the soft tissue borders prior to the removal of the incisors.

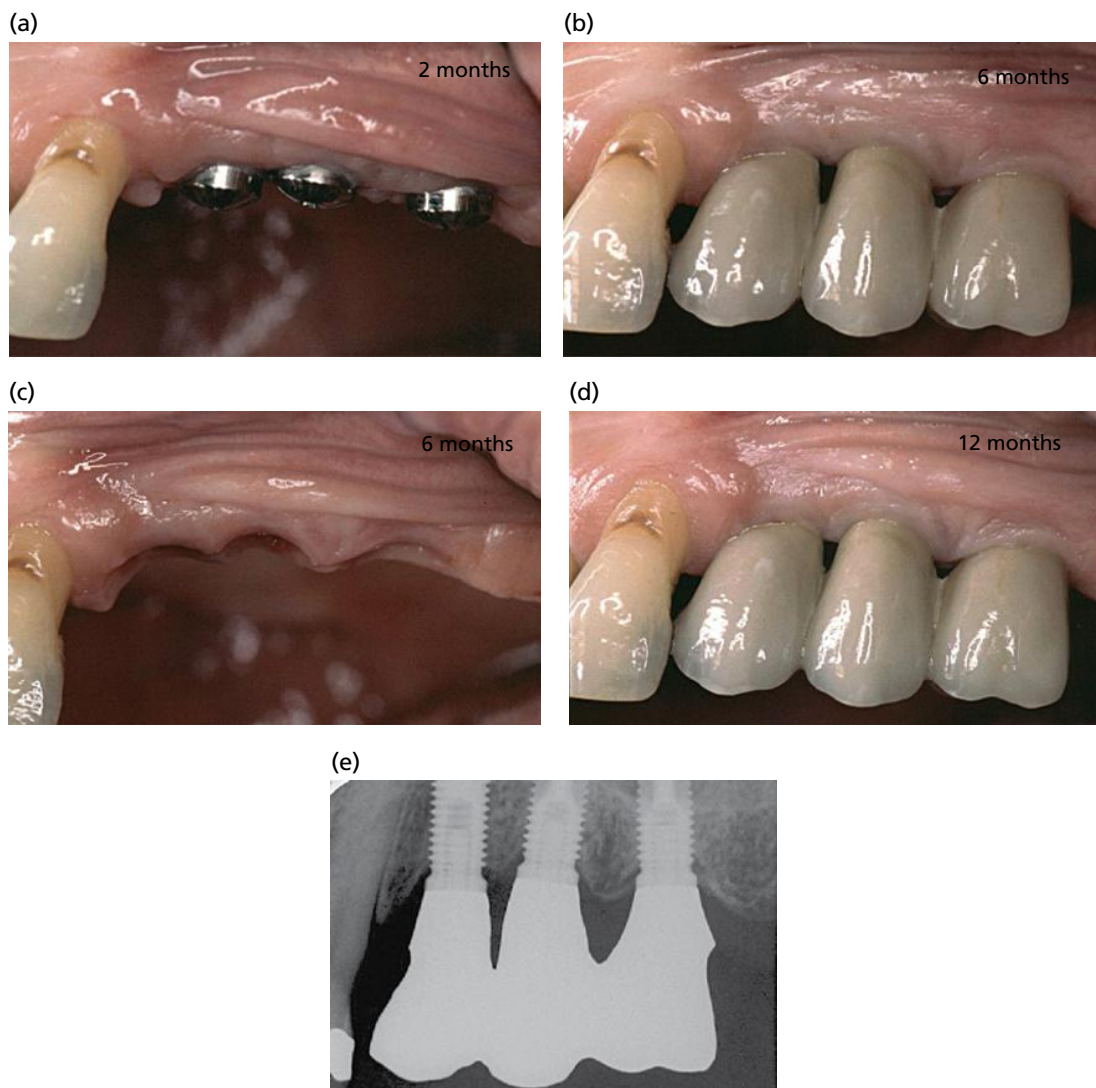


Fig. 4-37 See text for details.

restorations, the topography of the bone crest and the thickness of the supracrestal soft tissue portion are the factors that determine the position of the soft tissue margin in the interimplant area ("implant papilla"). Tarnow *et al.* (2003) assessed the height above the bone crest of the interimplant soft tissue ("implant papilla") by transmucosal probing at 136 anterior and posterior sites in 33 patients who had maintained implant-supported prostheses for at least 2 months. It was found that the mean height of the "papillae" was 3.4 mm, with 90% of the measurements in the range of 2–4 mm.

The dimension of the soft tissues between adjacent implants seems to be independent of the implant design. Lee *et al.* (2006) examined the soft tissue height between implants of two different systems (Brånemark Implant® and AstraTech Implant® systems), as well as the potential influence of the horizontal distance between implants. The height of the interimplant "papilla", that is the height of soft tissue coronal to the bone crest measured on radiographs, was about 3.1 mm for both implant systems. No difference was found regarding the "papilla" height for either of the implant systems at sites with a horizontal distance between the implants of <3 mm and those with a distance of 3 mm or greater. Gastaldo *et al.* (2004) evaluated the presence or absence of "papilla" at 96 interimplant sites in 58 patients. It was reported that the "papilla" filled the entire space between the implants only when the distance from the bone crest to the base of the contact point between the crown restorations, assessed by sounding, was <4 mm. Thus, taken together, these observations indicate that the soft tissue between two implants will have a maximum height of 3–4 mm, and that the location of the contact point between the crown restorations in relation to the bone crest level determines whether a complete soft tissue fill will be obtained in the embrasure space between two implants (Fig. 4-37).

References

- Abrahamsson, I. & Soldini, C. (2006). Probe penetration in periodontal and peri-implant tissues: an experimental study in the beagle dog. *Clinical Oral Implants Research* **17**, 601–605.
- Abrahamsson, I., Berglundh, T., Wennström, J. & Lindhe, J. (1996). The peri-implant hard and soft tissues at different implant systems. A comparative study in the dog. *Clinical Oral Implants Research* **7**, 212–219.
- Abrahamsson, I., Berglundh, T., Glantz, P.O. & Lindhe, J. (1998). The mucosal attachment at different abutments. An experimental study in dogs. *Journal of Clinical Periodontology* **25**, 721–727.
- Abrahamsson, I., Zitzmann, N.U., Berglundh, T. *et al.* (2002). The mucosal attachment to titanium implants with different surface characteristics: an experimental study in dogs. *Journal of Clinical Periodontology* **29**, 448–455.
- Andlin-Sobocki, A. & Bodin, L. (1993). Dimensional alterations of the gingiva related to changes of facial/lingual tooth position in permanent anterior teeth of children. A 2-year longitudinal study. *Journal of Clinical Periodontology* **20**, 219–224.
- Armitage, G.C., Svanberg, G.K. & Loe, H. (1977). Microscopic evaluation of clinical measurements of connective tissue attachment levels. *Journal of Clinical Periodontology* **4**, 173–190.
- Becker, W., Ochenbein, C., Tibbets, L. & Becker, B.E. (1997). Alveolar bone anatomic profiles as measured from dry skulls. *Journal of Clinical Periodontology* **24**, 727–731.
- Berglundh, T. (1999). Soft tissue interface and response to microbial challenge. In: Lang, N.P., Lindhe, J. & Karring, T., eds. *Implant Dentistry. Proceedings from 3rd European Workshop on Periodontology*. Berlin: Quintessence, pp. 153–174.
- Berglundh, T. & Lindhe, J. (1996). Dimensions of the peri-implant mucosa. Biological width revisited. *Journal of Clinical Periodontology* **23**, 971–973.
- Berglundh, T., Lindhe, J., Ericsson, I. *et al.* (1991). The soft tissue barrier at implants and teeth. *Clinical Oral Implants Research* **2**, 81–90.
- Berglundh, T., Lindhe, J., Jonsson, K. & Ericsson, I. (1994). The topography of the vascular systems in the periodontal and peri-implant tissues dog. *Journal of Clinical Periodontology* **21**, 189–193.
- Berglundh, T., Abrahamsson, I., Welander, M., Lang, N.P. & Lindhe, J. (2007). Morphogenesis of the periimplant mucosa. An experimental study in dogs. *Clinical Oral Implants Research* **18**, 1–8.
- Buser, D., Weber, H.P. & Lang, N.P. (1990). Tissue integration of non-submerged implants. 1-year results of a prospective study on 100 ITI-hollow-cylinder and hollow-screw implants. *Clinical Oral Implants Research* **1**, 225–235.
- Buser, D., Weber, H.P., Donath, K. *et al.* (1992). Soft tissue reactions to non-submerged unloaded titanium implants in beagle dogs. *Journal of Periodontology* **63**, 226–236.
- Chang, M., Wennström, J., Ödman, P. & Andersson, B. (1999). Implant supported single-tooth replacements compared to contralateral natural teeth. *Clinical Oral Implants Research* **10**, 185–194.
- Choquet, V., Hermans, M., Adriaenssens, P. *et al.* (2001). Clinical and radiographic evaluation of the papilla level adjacent to single-tooth dental implants. A retrospective study in the maxillary anterior region. *Journal of Periodontology* **72**, 1364–1371.
- Coatoam, G.W., Behrents, R.G. & Bissada, N.F. (1981). The width of keratinized gingiva during orthodontic treatment: its significance and impact on periodontal status. *Journal of Periodontology* **52**, 307–313.
- Ericsson, I. & Lindhe, J. (1993). Probing depth at implants and teeth. *Journal of Clinical Periodontology* **20**, 623–627.
- Gargiulo, A.W., Wentz, F.M. & Orban, B. (1961). Dimensions and relations of the dentogingival junction in humans. *Journal of Periodontology* **32**, 261–267.
- Gastaldo, J.F., Cury, P.R. & Sendyk, W.R. (2004). Effect of the vertical and horizontal distances between adjacent implants and between a tooth and an implant on the incidence of interproximal papilla. *Journal of Periodontology* **75**, 1242–1246.
- Gottlieb, B. (1921). Der Epithelansatz am Zahne. *Deutsche monatschrift führ Zahnheilkunde* **39**, 142–147.
- Gould, T.R.L., Westbury, L. & Brunette, D.M. (1984). Ultrastructural study of the attachment of human gingiva to titanium in vivo. *Journal of Prosthetic Dentistry* **52**, 418–420.
- Jemt, T. (1999). Restoring the gingival contour by means of provisional resin crowns after single-implant treatment. *International Journal of Periodontics and Restorative Dentistry* **19**, 21–29.
- Kan, J., Rungcharassaeng, K., Umez, K. & Kois, J. (2003). Dimensions of the periimplant mucosa: An evaluation of maxillary anterior single implants in humans. *Journal of Periodontology* **74**, 557–562.
- Kohl, J. & Zander, H. (1961). Morphology of interdental gingival tissue. *Oral Surgery, Oral Medicine, Oral Pathology* **60**, 287–295.
- Lang, N.P., Wetzel, A.C., Stich, H. & Caffesse, R.G. (1994). Histologic probe penetration in healthy and inflamed peri-implant tissues. *Clinical Oral Implants Research* **5**, 191–201.

- Lee, D-W., Park, K-H. & Moon, I-S. (2006). Dimension of interproximal soft tissue between adjacent implants in two distinctive implant systems. *Journal of Periodontology* **77**, 1080–1084.
- Magnusson, I. & Listgarten, M.A. (1980). Histological evaluation of probing depth following periodontal treatment. *Journal of Clinical Periodontology* **7**, 26–31.
- Matherson, D. & Zander, H. (1963). Evaluation of osseous surgery in monkeys. *Journal of Dental Research* **42**, 116.
- Mombelli, A., Mühle, T., Brägger, U., Lang, N.P. & Bürgin, W.B. (1997). Comparison of periodontal and peri-implant probing by depth-force pattern analysis. *Clinical Oral Implants Research* **8**, 448–454.
- Moon, I-S., Berglundh, T., Abrahamsson, I., Linder, E. & Lindhe, J. (1999). The barrier between the keratinized mucosa and the dental implant. An experimental study in the dog. *Journal of Clinical Periodontology* **26**, 658–663.
- Müller, H.P. & Könönen, E. (2005). Variance components of gingival thickness. *Journal of Periodontal Research* **40**, 239–244.
- O'Connor, T.W. & Biggs, N. (1964). Interproximal craters. *Journal of Periodontology* **35**, 326–330.
- Olsson, M. & Lindhe, J. (1991). Periodontal characteristics in individuals with varying forms of upper central incisors. *Journal of Clinical Periodontology* **18**, 78–82.
- Olsson, M., Lindhe, J. & Marinello, C. (1993). On the relationship between crown form and clinical features of the gingiva in adolescents. *Journal of Clinical Periodontology* **20**, 570–577.
- Orban, B. & Köhler, J. (1924). Die physiologische Zahnfleischetasche, Epithelansatz und Epitheltiefenwucherung. *Zeitschrift für Stomatologie* **22**, 353.
- Oschnebein, C. & Ross, S. (1969). A reevaluation of osseous surgery. In: *Dental Clinics of North America*. Philadelphia, PA: W.B. Saunders, pp. 87–102.
- Pontoriero, R. & Carnevale, G. (2001). Surgical crown lengthening: A 12-month clinical wound healing study. *Journal of Periodontology* **72**, 841–848.
- Quirynen, M., van Steenberghe, D., Jacobs, R., Schotte, A. & Darius, P. (1991). The reliability of pocket probing around screw-type implants. *Clinical Oral Implants Research* **2**, 186–192.
- Ryser, M.R., Block, M.S. & Mercante, D.E. (2005). Correlation of papilla to crestal bone levels around single tooth implants in immediate or delayed crown protocols. *Journal of Maxillofacial Surgery* **63**, 1184–1195.
- Schou, S., Holmstrup, P., Stolze, K. *et al.* (2002). Probing around implants and teeth with healthy or inflamed marginal tissues. A histologic comparison in cynomolgus monkeys (*Macaca fascicularis*). *Clinical Oral Implants Research* **13**, 113–126.
- Schropp, L., Wenzel, A., Kostopoulos, L. & Karring, T. (2003). Bone healing and soft tissue contour changes following single-tooth extraction: A clinical and radiographic 12-month prospective study. *International Journal of Periodontics and Restorative Dentistry* **23**, 313–323.
- Sicher, H. (1959). Changing concepts of the supporting dental structure. *Oral Surgery, Oral Medicine, Oral Pathology* **12**, 31–35.
- Tarnow, D., Magner, A. & Fletcher, P. (1992). The effect of the distance from the contact point to the crest of bone on the presence or absence of the interproximal dental papilla. *Journal of Periodontology* **63**, 995–996.
- Tarnow, D., Elian, N., Fletcher, P. *et al.* (2003). Vertical distance from the crest of bone to the height of the interproximal papilla between adjacent implants. *Journal of Periodontology* **74**, 1785–1788.
- van der Velden, U. (1982). Regeneration of the interdental soft tissues following denudation procedures. *Journal of Clinical Periodontology* **9**, 455–459.
- Waerhaug, J. (1952). Gingival pocket: anatomy, pathology, deepening and elimination. *Odontologisk Tidskrift* **60** (Suppl 1).
- Weisgold, A. (1977). Contours of the full crown restoration. *Alpha Omegan* **7**, 77–89.
- Welander, M., Abrahamsson, I. & Berglundh, T. (2008). The mucosal barrier at implant abutments of different materials. An experimental study in dogs. *Clinical Oral Implants Research* **19**, 635–641.
- Wheeler, R.C. (1961). Complete crown form and the periodontium. *Journal of Prosthetic Dentistry* **11**, 722–734.
- Wheeler, R.C. (1966). *An Atlas of Tooth Form*. Philadelphia: W.B. Saunders Co, pp. 24–25.